

Monsij Biswal

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EDUCATION

University of California, Santa Barbara, USA

MS/PhD, Electrical and Computer Engineering

October 2020 – Present

Major: Communications Minor: Signal Processing

National Institute of Technology, Durgapur, IN

B.Tech, Electronics and Communication Engineering

July 2016 – June 2020

PUBLICATIONS

- **M. Biswal**, A. Chandra, A. U. Rahman et al., "On the Characterization of Beam Misalignment in Outdoor-to-Indoor 60 GHz mmWave Channel", 15th European Conference on Antennas and Propagation (EuCAP), Dusseldorf, Germany, 2021
 - A. Sen, **M. Biswal**, S. Datta "Intelligent Traffic Routing Based on Real-time Congestion Analysis", 16th IEEE India Council Conference, Gujarat, India, 2019
 - A. U. Rahman, **M. Biswal**, "Error-tolerant Beam Steering of mmWave Antennas by Trajectory Estimation of Highway Vehicles", 11th International Conference on Communication Systems & Networks (COMSNETS), Bengaluru, India, 2019
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WORK EXPERIENCE

Christian Albrechts Universität zu Kiel, Germany

Research Intern, DAAD WISE Scholar

May 2019 – July 2019

Indian Institute of Technology, Kharagpur, India

Summer Intern

May 2018 – June 2018

PROJECTS

Millimeter Wave Channel Modelling and Performance Indicators

Mentor: Associate Prof. Aniruddha Chandra

August 2019 – Present

National Institute of Technology Durgapur

- Channel modelling based on channel sounding measurements in a vehicular & outdoor scenario centered around a specific license free band of 60GHz.
- Extensive analysis of small scale and large scale fading parameters and their effects on rms delay spread of the channel.

Custom Neural Network Training Function for Optical Transmission System

Mentor: Prof. Dr.-Ing. Stephan Pachnicke

May 2019 – July 2019

Christian Albrechts Universität zu Kiel

- Analyzed various neural network training algorithms for equalizing a higher order modulation format in a optical transmission system.
- Implemented a custom training function to improve the accuracy of the constellation diagram.

Interactive GUI for ROI Selection of Medical Images

Mentor: Prof. Sudipta Mukhopadhyay

December 2018 – January 2019

Indian Institute of Technology, Kharagpur

- Framed a GUI in MATLAB for interactive selection of ROI from frontal image samples of mouth cancer victims.
- Classified an image as malignant or benign based on the output of image processing algorithm.

Data Regeneration codes for distributed storage systems

Mentor: Prof. Sant Sharan Pathak

May 2018 – July 2018

Indian Institute of Technology, Kharagpur

- Implemented and tested various efficient storage algorithms to control data reliability and redundancy.
- Developed regenerating codes, a class of codes to prevent errors during data reconstruction and retrieval in a distributed storage system.

HONOURS AND AWARDS

- **DAAD WISE Scholarship Recipient** *January 2019*
Was one among 80 students selected for a funded summer internship at Germany
- **MITACS Globalink Research Intern Award** *January 2019*
Invited to participate in a research project at Laval University, Quebec
- **16th KVS Junior Mathematics Olympiad Awardee** *October 2014*
For being among the top 30 candidates in India
Invited to attend training camp for Indian National Mathematics Olympiad (INMO)

SKILLS & OTHERS

Programming: Python, C/C++, Julia

Softwares: MATLAB, CorelDRAW

RELEVANT COURSES

ECE–240: Optimal Estimation and Filtering

ECE–250: Wireless Communication and Networks

EC–601: Transmission Lines

EC–503: Digital Signal Processing

EE–431: Control Systems Engineering

ECE–284: Theoretical Machine Learning

ECE–235: Stochastic Processes

EC-501: Digital Communication

EC–504: Computer Architecture and Organization

EC–402: Analog Communication

POSITIONS OF RESPONSIBILITY

Secretary, IEEE Student Branch, NIT Durgapur

April 2019 – March 2020

Being a direct handshake with IEEE, IEEE Student Branch aims to promote research related activities among the youths of the campus. These include events such as seminars, workshops, guest lectures and internship bootcamps.

- Organized short-term course on MATLAB and various workshops for autonomous robotics.
- Collaborated with various other student branches for organising national level events.

Representative, Dept. of Training and Placement, NIT Durgapur

February 2018 – March 2020

- Managed end-to-end recruitment process of various companies visiting the campus including logistics and hospitality.
 - Ensured proper exchange of information between students and the business executives.
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